TANG XINYI (唐鑫夷)

Email: tangxy2001@gmail.com | **GitHub:** github.com/tangxy2001-sketch
Personal Website: https://www.xinyitang.me/

EDUCATION

Beijing Normal University, Beijing, China 2023 – 2026
Master of Arts, School of Arts and Media (GPA: 3.8/4.0)
Supervisor: Prof. Lun Zhang
Thesis: Imagining the Media Roles of Conversational AI: A Computational Grounded Theory Study with Multi-Agent Frameworks

Communication University of China, Beijing, China 2019 – 2023
Bachelor of Engineering, School of Information and Communication Engineering (GPA: 3.65/4.0)
Academic supervisor in the Excellence Program: Prof. Fulian Yin
Thesis: Dynamics of information propagation and intervention in the multi-platform coupled networks

Summer Institute in Computational Social Science, Singapore 2026
Participant, National University of Singapore

HONORS & AWARDS

Master's Thesis Recognized as Outstanding, Beijing Normal University 2026

Third Prize, Outstanding Paper Award 2026
2nd International Research Workshop on Human-Machine Communication, Psychology, and Social Change

First-Class Academic Scholarship, Beijing Normal University 2025

Second-Class Academic Scholarship, Beijing Normal University 2024

Outstanding Graduate in Beijing, China 2023

Outstanding Graduate at Communication University of China 2023

Second Prize, National Mathematical Modeling Contest (Beijing Division) 2021

Multiple scholarships, Communication University of China 2019 – 2023
Including First-Class Academic Scholarship, Outstanding Student Leader Scholarship, etc.

RESEARCH INTERESTS

AI for Families, Family Relationships, and Human-Centered AI Design

• My research examines the role of AI in family life through an integrated research-and-design approach.
• First, I investigate how AI technologies shape interpersonal relationships within families—particularly couple and parent – child relationships—and the social and cultural mechanisms underlying these changes.
• Second, I aim to translate research insights into the design and development of family-oriented AI products that support more equitable divisions of household and caregiving labor, improve family coordination, and facilitate collaboration among family members.

Social Computing & Public Opinion

• My research uses computational methods to examine information diffusion and public opinion dynamics on social media platforms.
• I am particularly interested in the structural patterns of information flow, the formation and evolution of opinions, and the emergence of polarization in online networks.
• Methodologically, I draw on social network analysis, topic modeling, machine learning, and large-scale social media data analysis.

ACADEMIC SERVICE

Teaching Assistant: *Vibe Coding with AI*, Beijing Normal University (2024 – 2025)

Peer Reviewer: *Information Processing & Management* (SCI TOP), *Physica D: Nonlinear Phenomena (SCIE), ICA 2025 Annual Conference (Human-Machine Communication Interest Group)*

RESEARCH EXPERIENCE

Human-AI Relational Bonding in Chinese Social Structures 2023 – 2025

Principal Investigator

- Investigated how interaction with conversational AI discloses pre-existing relational tensions in Chinese society, engaging the theory of **the differential mode of association (chaxu geju)** to trace the cultural antecedents, process mechanisms, and behavioral outcomes of human-AI relational bonding.
- Developed **Media Compensation Theory** as a theoretical contribution to HCI, accounting for how AI comes to compensate for unmet relational needs.
- Designed an **LLM-based multi-agent framework** in which agents hold distinct epistemic roles to conduct computational grounded theory, externalizing methodological commitments as auditable constraints.
- **Awarded Third Prize for Outstanding Paper**, 2nd International Research Workshop on Human-Machine Communication, Psychology, and Social Change; manuscript in preparation for CHI 2027.

Public Opinion Dynamics: Social Simulation and Intervention 2021 – 2023

Project Leader, National Innovation and Entrepreneurship Training Program

- Built complex-network models simulating public opinion dynamics on social media.
- Designed counterfactual intervention strategies for rumor mitigation and affective contagion during public health emergencies.
- Output: three national invention patents and seven peer-reviewed publications.

SELECTED PRODUCT & DESIGN PROJECTS

NearbyLove × CommunitySense 2026

Product Manager

Live Demo: nearbylove.github.io/nearby_demo

- Developed a community-oriented AI service system grounded in anthropologist Biao Xiang's concept of **"the nearby"** and his critique of its disappearance in contemporary urban life.
- Translated research on weakened neighborhood ties, solo living, mobile populations, and community governance into a multi-stakeholder product connecting residents, property managers, neighborhood committees, and local governments.
- Designed and built an interactive prototype that uses AI-assisted coordination and privacy-preserving public-space sensing to support mutual aid, collective action, and everyday community connection.

INDUSTRY & ENGINEERING EXPERIENCE

China Telecom System Integration Co., Ltd. 2023

Product Manager Intern

- Designed product prototypes and authored requirement documents for social-computing systems deployed in real-world governance contexts (judicial and public-security agencies), translating research-grade methods on opinion dynamics into operational tools for institutional users.

National Key R&D Program of China (No. 2021YFF0901705) 2021 – 2023

Core R&D Member · Sub-project: Cultural Resource Reuse and Big Data Services

- Built end-to-end data pipelines (web crawling, cleaning, mining, analysis) and rapid product demos for big-data applications in the cultural production industry, working at the engineering layer rather than the research layer.

PUBLICATIONS

Conference Papers

[1] **Tang, X. Y**. (2026, May). Imagining the Media Roles of Generative AI: A Computational Grounded Theory Study Using Large Language Models. *2nd International Research Workshop on Human-Machine Communication, Psychology, and Social Change,* Shenzhen, China. **Third Prize, Outstanding Paper Award.** English version in preparation for CHI 2027.

[2] **Tang, X. Y.**, Lu, J. L., Zhang, L., Wu, Y., & Xu, X. K. (2025). Aligning AI With What: Role Coordination through Perception, Construction, and Adaptation during Human-AI Interactions. Paper presented at the Human-Machine Communication Division, *2025 ICA Annual Conference.* Denver, USA.

[3] Wang, J. X., Liang, T. Y., Kuang, Q. H., **Tang, X. Y.**, Ma, R., & Yin, F. L. (2022). Combined influence of commenting and forwarding on information propagation on the Chinese Sina Microblog. *2022 7th International Conference on Cloud Computing and Big Data Analytics*, Chengdu, China, pp. 342–347.

Journal Articles

[1] **Tang, X. Y.**, & Zhang, L. (2026). Paradigmatic Integration, Consolidation, and Differentiation: Information Infrastructure and the Paradigm Shifts of Journalism and Communication Studies. *Journalism & Communication Studies*. **(Under review, final round, CSSCI TOP)**

[2] Yin, F. L., Wang, J. X., **Tang, X. Y.**, Jiang, X. Y., Wu, Y. W., & Wu, J. H. (2025). Dynamics of information propagation and intervention in the multiplatform coupled networks. *Expert Systems with Applications*, 277, 127262. **(SCI TOP)**

[3] Yin, F. L., **Tang, X. Y.**, Liang, T. Y., Kuang, Q. H., Wang, J. X., Ma, R., ... & Wu, J. H. (2024). Coupled dynamics of information propagation and emotion influence: Emerging emotion clusters for public health emergency messages on the Chinese Sina Microblog. *Physica A: Statistical Mechanics and its Applications*, 639, 129630. **(SCI Q2)**

[4] Yin, F. L., She, Y. W., Pan, Y. Y., **Tang, X. Y.**, Hou, H. T., & Wu, J. H. (2023). Hot-topics cross-propagation and opinion-transfer dynamics in the Chinese Sina-microblog social media: A modeling study. *Journal of Theoretical Biology*, 566, 111480. **(SCI Q3)**

[5] Yin, F. L., **Tang, X. Y.**, Liang, T. Y., Huang, Y. J., & Wu, J. H. (2022). External intervention model with direct and indirect propagation behaviors on social media platforms. *Mathematical Biosciences and Engineering*, 19(11), 11380–11398. **(SCI Q2)**

[6] Yin, F. L., Pan, Y. Y., **Tang, X. Y.**, Wu, C., Jin, Z., & Wu, J. H. (2022). An information propagation network dynamic considering multi-platform influences. *Applied Mathematics Letters*, 133, 108231. **(SCI TOP)**

[7] Yin, F. L., Wu, Z. L., Shao, X. Y., **Tang, X. Y.**, Liang, T. Y., & Wu, J. H. (2022). Topic—a cluster of relevant messages—propagation dynamics: A modeling study of the impact of user repeated forwarding behaviors. *Applied Mathematics Letters*, 127, 107819. **(SCI TOP)**

Patents

[1] Information propagation analysis method and system based on social media pseudo-environment modeling. National Invention Patent No. ZL202210645833.8. Granted 2022.

[2] Hybrid information propagation dynamics model and its information propagation analysis method. National Invention Patent No. ZL202210218560.9. Granted 2022.

[3] Public topic propagation evaluation method and system based on modeling. National Invention Patent No. ZL202110812031.7. Granted 2021.

RELEVANT RESEARCH SKILLS

Qualitative Methods: Constructivist grounded theory, semi-structured interviewing

Computational Methods: LLM-based multi-agent system design, NLP

Programming & Tools: Python, R, Vibe Coding